

DRY-RUN GUIDE | billing_cart.js

Source: javascript\2nd_belt\billing_cart.js

CORE BEHAVIOR

canSellCookie checks whether each customer can be served exact change: retain \$5 bills; a \$10 consumes one \$5; a \$20 prefers one \$10 plus one \$5, otherwise three \$5. The script prints results for four fixed queues.

REPRESENTATIVE INPUT / CALL

Built-in queues: [5,5,5,10,20], [5,5,10,10,20], [5,5,5,10,10,20], [5,5,5,5,20].

STEP TRACE

Queue 1 succeeds (20 uses 10+5); queue 2 fails at its second 10; queue 3 fails at 20 with only two 5s; queue 4 succeeds using three 5s for 20.

EXPECTED OUTPUT / RETURN

Four lines: 'true', 'false', 'false', 'true'.

ACTUAL SOURCE EXCERPT

Verbatim lines from the source; long lines wrap. Source files were not modified.

```
001 | // Question: Calculate the total bill for the items in a shopping cart.
```

```
002 | // Add your solution here.
```

```
003 |
```

```
004 | function canSellCookie(bills) {
```

```
005 |     let five = 0;
```

```
006 |     let ten = 0;
```

```
007 |     let twenty = 0
```

```
008 |
```

```
009 |     for (let i = 0; i < bills.length; i++) {
```

```
010 |         const bill = bills[i];
```

```
011 |
```

```
012 |         if (bill === 5) {
```

```
013 |             five++;
```

```
014 |         } else if (bill === 10) {
```

```
015 |             if (five === 0) return false;
```

```
016 |             five--;
```

```
017 |             ten++;
```

```
018 |         } else if (bill === 20) {
```

```
019 |             if (ten > 0 && five > 0) {
```

```
020 |                 ten--;
```

```
021 |                 five--;
```

```
022 |                 twenty++;
```

```
023 |             } else if (five >= 3) {
```

```
024 |                 five -= 3;
```

```
025 |                 twenty++;
```

```
026 |             } else {
```

```
027 |                 return false;
```

```
028 |             }
```

```
029 |         }
```

```
030 |     }
```

```
031 |
```

```
032 |     return true;
```

```
033 | }
```

```
034 |
```

```
035 | console.log(canSellCookie([5, 5, 5, 10, 20]));
```

```
036 | console.log(canSellCookie([5, 5, 10, 10, 20]));  
037 | console.log(canSellCookie([5, 5, 5, 10, 10, 20]));  
038 | console.log(canSellCookie([5, 5, 5, 5, 20]));  
039 |
```